ASSESSMENT TOOL 2 – CASE STUDY

Course: Computer Vision (ITA05) | CO3 – Precision (BL4)

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Q1. Satellite Image Registration – Edge Detection vs Harris Corner Detection

Limitations of Edge Detection in This Application

Edge detection identifies boundaries by locating sharp intensity gradients in an image, but it does not distinguish between different types of edges or guarantee that the same physical point will be detected consistently across two images. In satellite image registration, this creates several problems:

- Illumination sensitivity: Since the two images are captured at different times, sun angle, shadows, and seasonal lighting change the intensity gradients, so edges that appear strong in one image may be weak or missing in the other.
- Orientation and viewpoint changes: Edge maps are not inherently rotation- or scale-invariant, so a boundary detected in one orientation may not align with the same boundary rotated in the second image.
- Ambiguous correspondence: Edges are typically long, continuous curves (e.g., roads, coastlines, field boundaries). Any point along a straight or smoothly curved edge looks similar to its neighbors, making it hard to establish a unique, reliable match between two images — this is known as the aperture problem.
- Background clutter: Natural and urban backgrounds introduce many irrelevant edges (vegetation texture, cloud boundaries, noise), which increases false matches and reduces alignment accuracy.

How Harris Corner Detection Improves the Process

Harris Corner Detection identifies corner points — locations where intensity changes sharply in two orthogonal directions simultaneously — rather than following a single edge direction. This directly resolves the shortcomings of edge-based matching:

- Distinctive localization: Corners are point-like and highly localized (e.g., building corners, road intersections, field boundary junctions), which removes the ambiguity of matching along a continuous edge and gives a precise, repeatable location.
- Improved matching accuracy: Because corners have strong gradients in two directions, they can be uniquely identified and matched between the two images with far fewer false correspondences than plain edge points.
- Robustness to illumination: The Harris response is based on local gradient structure (via the second-moment matrix), which remains comparatively stable under moderate illumination changes, since corner geometry does not change even if brightness does.
- Rotation invariance: The Harris corner response is mathematically invariant to image rotation, so corners remain detectable even when the satellite images differ in orientation.
- Reduced sensitivity to background clutter: Only locations with strong two-directional gradients qualify as corners, filtering out much of the noisy, low-information background that misleads edge-based matching.

Conclusion

Replacing edge detection with Harris Corner Detection allows the registration system to rely on a sparse set of distinctive, repeatable feature points instead of ambiguous continuous edges. This significantly improves feature localization accuracy, reduces mismatches caused by illumination and orientation differences, and produces a more robust and reliable alignment of satellite images captured under varying imaging conditions.

Q2. Image Representation for Real-Time Object Detection

Impact of Image Representation on Computational Efficiency

A standard color image stores three channels (Red, Green, Blue) per pixel, whereas a grayscale image stores a single intensity channel, and a binary image stores just one bit per pixel. Since feature extraction and detection algorithms process every channel of every pixel, working directly on color images roughly triples the amount of data that must be processed compared to grayscale, directly increasing computation time and memory usage. This additional overhead is a major reason the surveillance system in the case struggles to keep up with real-time requirements.

Impact on Edge Detection

Edge detection algorithms (e.g., Sobel, Canny) operate on intensity gradients. Converting to grayscale collapses the three color channels into one intensity value, which is sufficient for most edge detection tasks since object boundaries are primarily defined by changes in brightness rather than hue. This means edge detection accuracy is largely preserved on grayscale images while the computation required is reduced substantially. Binary images, on the other hand, lose fine gradient information entirely — edges become simple foreground/background transitions, which speeds up processing further but can discard subtle boundaries and reduce detection accuracy in complex scenes.

Impact on Feature Extraction Performance

Most classical feature detectors (Harris corners, SIFT, HOG) are designed to operate on grayscale intensity information, since geometric structure (corners, gradients, textures) is captured effectively without color. Using grayscale therefore does not significantly degrade feature extraction quality while greatly lowering computational cost. Binary representations, however, remove intensity gradation almost completely, which can cause loss of texture and weak-contrast features, making them unsuitable for extracting rich, reliable features needed to distinguish between similar objects (e.g., a pedestrian vs a pole).

Evaluation & Justification

Grayscale representation offers the best trade-off for this application. It reduces computational load by roughly two-thirds compared to color, directly improving processing speed and helping meet real-time requirements, while retaining the intensity gradient information that edge detection and feature extraction algorithms depend on. Binary images process even faster but at the cost of losing critical shape and texture detail, which risks missing or misclassifying vehicles and pedestrians — an unacceptable trade-off for a safety-relevant surveillance task. Full color should only be retained if the application specifically requires color-based cues (e.g., traffic light or brake-light detection); otherwise, converting to grayscale before feature detection is the most appropriate and efficient choice.

Conclusion

For accurate and reliable real-time object detection, grayscale image representation is recommended as it substantially improves computational efficiency while preserving the structural information required for robust edge detection and feature extraction, striking the best balance between speed and accuracy for the autonomous surveillance system.